



Ancient crater lake on mars

Physicists at Brown University have discovered a brand new ancient crater lake on Mars. <http://dgit.in/apr21-59>



NASA's psyche

A major component of NASA's Psyche spacecraft has allowed its final assembly to continue. <http://dgit.in/apr21-60>

How ESA Tech Is Changing Our World

The real-world applications of ESA's space technology.

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OVERVIEW

Space 'spin-offs' are technologies, products and processes developed by space agencies around the world, to use in space – but they are now used as terrestrial technology (on Earth). Today we take a closer look at the European Space Agency's (ESA) Technology, and how they are being used in creative ways here on Earth.

WATER PURIFIERS

Water is the elixir of life, and without it we wouldn't survive very long. It is by far one of the most consumed items in the world along with also being some of the heaviest items that are stored aboard a manned spacecraft.

This particularly becomes a problem, on missions of long durations, where the crew requires large amounts of water to consume. To help reduce the amount of water that needs to be taken aboard, ESA's Space Research and Technology Centre (ESTEC) in Holland developed an automated filtration system to recycle waste water into drinking water. This includes the condensation that forms on the surfaces inside the spacecraft, and the water discharged from washing clothes, dishes, experiments and the life-support system! What makes this challenging, is that recycling water from a different number of sources would often lead to difficulties in anticipating what impurities may be present. For example, it could contain volatile organic compounds (VOCs) and pathogenic microbes which are both extremely difficult to get rid of. This is



where the system developed by ESA shines. The system has proven to be extremely effective at removing all types of contamination. The way it works is, there are a series of membranes which filter out the various impurities from the waste water. This system is capable of treating hundreds of litres of 'grey' water per

hour! The success of this tech has attracted the interest of a major European bottler of spring water. The company was looking for a filtration system to deal with the problems faced at several water sources. Water extracted from some natural springs can be discoloured due to the high concentration of minerals in them, and after conducting trials, the ESA filtration system proved to be highly efficient at removing these minerals and other impurities. This system is now also being considered for recycling waste on ocean cruise liners!